Executive Summary: This week the team had our PDR with Mr.Nooner in which we laid out the plan for our project with Dr.Fitzmorris and the team with Mr.Nooner. After taking considerations from Dr.Fitzmorris we changed the layout of the software block diagram slightly to make the button still work as intended but now when the audio for a gunshot is detected that will also trigger the same things that the button would as well. We also researched and finished some implementation of the bluetooth on our device. This week our Hardware team also created and finished their individual schematics for all their components and next week we have plans to have a completed full schematic of the entire board. We also started the creation of the CDR and will have that ready for the future data planned at the end of March. This upcoming week the team will also be in contact with Mr.Nooners team for any follow up questions after all PDR's have been completed.

Hardware:

Cody C.

Progress:

- Practice PDR with the group (~1 Hour)
- 3/2 – Preliminary Design Review (~1 Hour)
- Schematic:
    - Modify Arduino Nano ESP32 Schematics (~ 30 minutes)
    - Create Schematic for I2S MEMS Microphone (~ 15 minutes)
    - Create Schematic for Boost Converter (~ 15 minutes)
    - Create Schematic for Button (~ < 2 Minutes)
- Create rough schematics for all requirements for the gunshot detection (~ 1 Hour)
    - Found an updated data sheet for the I2S MEMS Microphone

- Researched how to connect a boost converter with Arduino Nano ESP32 (~ 40 Minutes)
    - Research what components I need to convert 3.7V to 5V
        - Two resistors are needed to set the desired voltage to 5V
        - Inductor, Diode, and Capacitor are needed for the new Vout voltage
- Upload DFRobot resources and KiCad Schematics to GitHub (~20 Minutes)
- 3/8 – Weekly meeting with the team (~ 20 Minutes)

Total Hours Worked: ~ 5 Hours 22 Minutes

Future Work/Plan

- Create combine hardware schematic
    - Battery, Boost Converter, GPS Module, Gunshot Detection, and Arduino Nano ESP32

Hardware:

Ryan K.

Progress:

- 3/2 (2hr)
    - Preliminary Design Review practice and presentation
- 3/4 (36min)
    - Created remaining custom components in KiCad for schematic
- 3/6 (3h 28min)
    - Reviewed component datasheets for specifications and limitations

    - Created rough draft schematic of the project (excluding the microphone)

Total Hours Worked: 6h 4min

Future Work/Plan:

- Update and finalize overall schematic

Software:

Alex H.

Progress:

- 3/4 (2 h 7 min)
    - Set up initial software files for the project and pushed them to the *software* branch of the GitHub repository:
        - WearableAlertDevice.ino
        - GPSModule.h/.cpp
        - GunshotDetection.h/.cpp
        - WirelessDataTransmission.h/.cpp
        - config.h
    - Modified the software compiler in Arduino IDE to detect GPIO numbers instead of regular pin numbers (gives us more ability to control the ESP32).
    - Implemented the push button functionality in the software using an interrupt function.
    - Read datasheet for microphone module to determine how to interpret its digital output data as audio in the software.
- 3/6 (43 min)

    - Tested the push button's functionality by wiring a simple button circuit on a breadboard and the ESP32 microcontroller.
    - Successfully implemented the push button functionality and reduced debouncing on the button.
- 3/7 (2 h 12 min)
    - Worked on implementing Bluetooth connectivity in software to transmit data.
    - Found a mobile app called nRF Connect by Nordic that can connect to ESP32 microcontrollers and read data sent from the microcontroller.
    - Able to connect Wearable Bluetooth Device to mobile app on phone, but data does not appear on mobile app (unsure if this is due to settings on phone or device software).

Total Hours Worked: 5 h 2 min

Future Work/Plan:

- Continue working on Bluetooth communication functionality and send artificial data to the mobile app on the phone.

Project Manager:

Jonas W.

Progress:

- 3/2 (2hr)
    - Preliminary Design Review practice and presentation
- 3/3 (45 min)
    - Gathered documents for data sheet booklet after PDR and new data sheets
- 3/6 (1 hr)

    - Creation of WAR and CDR Slides
    - Choosing of CDR theme and slide layout
- 3/7 (1 hr)
    - Started process of parts ordering with our budget sheet over to Dr. Fitzmorris for Monday.

Total Hours Worked: 4hr 45min

Future Work/Plan: Next week I plan on having the planning of our CDR completed and ready for the spring break gap that will be taking place the week after. We are hoping to have parts ordered as well so that we may begin testing and debugging after spring break when they hopefully arrive.